Example: Status after n clocks
Tomasulo with Reorder Buffer

Tomasulo with Reorder Example 1

Tomasulo with Reorder Buffer: Cycles 1 to 3 | Video 10c

https://www.youtube.com/watch?v=umeqKe0o6nA&list=WL&index=10&t=34s

https://www.youtube.com/watch?v=ihW5uuEVabE&list=PLNrZ57svi8FpW1ibXjM6gZnTN7UFGiMT3&index=20

Tomasulo with RoB

1. Issue and Read:
   - structural hazard detection for entry in reservation station & reorder buffer
     - issue if no hazard
     - stall if hazard
   - Read registers for source operands
     - can come from either registers or reorder buffer

2. Execute
   - Monitor the CDB (common data bus) for operands → operand if available is transferred to reservation station
     - Reservation station tag is entry number in reorder buffer
     - Handle RAW hazards

Dispatch

RS

CDB

Execute - ALU

Tomasulo with Reorder Buffer: Cycles 1 to 3 | Video 10c

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Tomasulo with RoB

Write result

- Broadcast result & reorder buffer entry (tag) on the common data bus to reservation stations, store buffer & reorder buffer

RoB ⇒ ARF

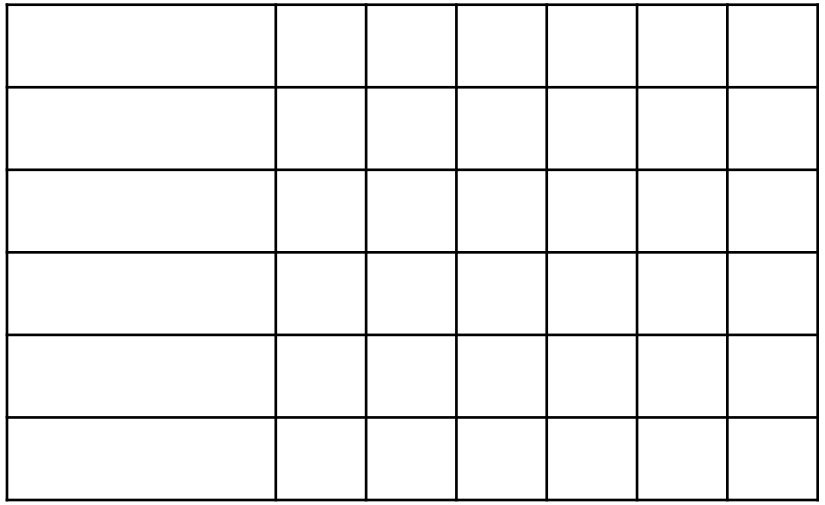
Commit

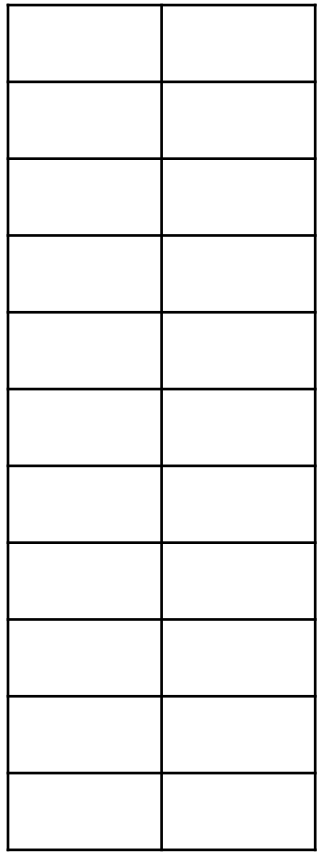
- Commit the instruction at the head of the reorder buffer
- Update register with result in reorder buffer
- Remove instruction from reorder buffer

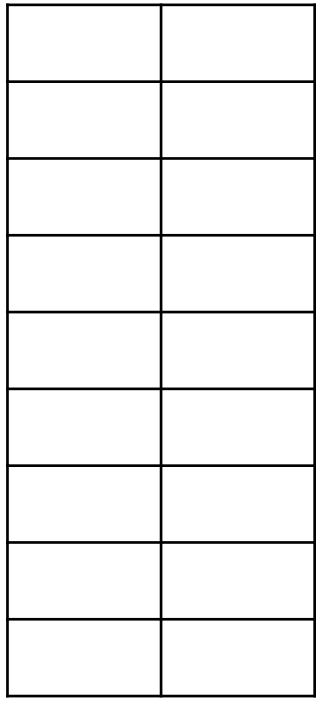


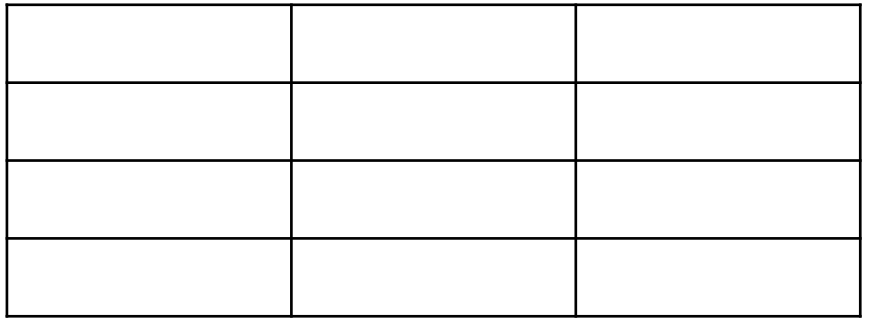
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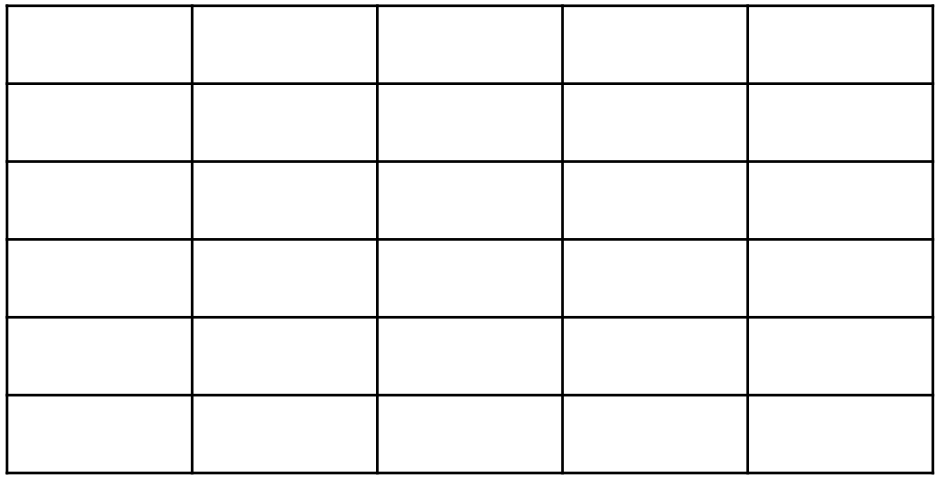


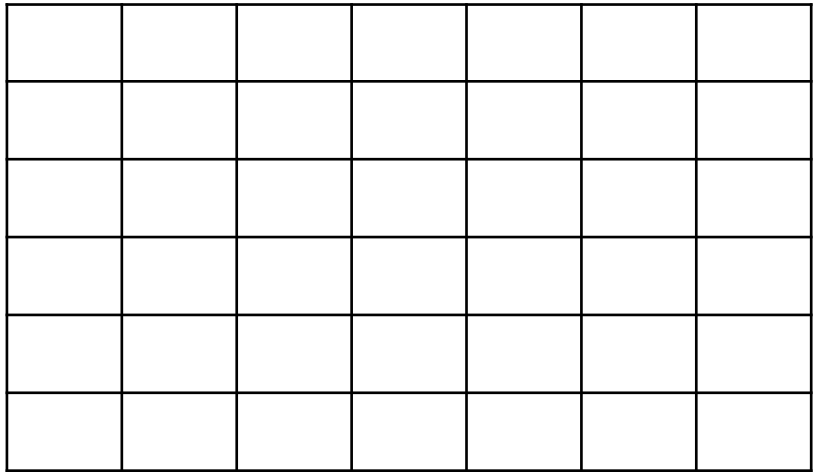






Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=20 Units Delay





Instr	ISU	EXST	EXCMP	WR	COM

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

Load Buffers / RS

	BUSY	**ADDR**
LOAD 1		
LOAD 2		
LOAD 3		

Architecture Features

ROB

	INST	**DEST**	**VAL**	
ROB1				
ROB2				
ROB3				
ROB4				

ARF

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	
R2	
R3	

RAT

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	

RS

TYPE	**BUSY**	**OP**	**S1**	**S2**	**W1**	**W2**

	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)					
LD F2, 45(R3)					
MUL F0, F2, F4					
SUB F8, F6, F2					
DIV F10, F0, F6					
ADD F6, F8, F2					

Load Buffers / RS

	BUSY	ADDR
LOAD 1		
LOAD 2		
LOAD 3		

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

ROB

	INST	DEST	VAL	
ROB1				
ROB2				
ROB3				
ROB4				

ARF

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	
R2	
R3	

RAT

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	

RS

TYPE	BUSY	OP	S1	S2	W1	W2

	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)					
LD F2, 45(R3)					
MUL F0, F2, F4					
SUB F8, F6, F2					
DIV F10, F0, F6					
ADD F6, F8, F2					

Showing Cycles

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

Load Buffers / RS

	BUSY	ADDR
LOAD 1		
LOAD 2		
LOAD 3		

ROB

	INST	DEST	VAL	
ROB1				
ROB2				
ROB3				
ROB4				
ROB5				
ROB6				

ARF

Architecture Register File

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	
R2	
R3	

RAT

Register Allocation Table

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1						
Add2						
Add3						
Mul1						
Mul2						

Dispatch →	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1				
LD F2, 45(R3)					
MUL F0, F2, F4					
SUB F8, F6, F2					
DIV F10, F0, F6					
ADD F6, F8, F2					

Showing Cycles

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

Load Buffers / RS

	BUSY	ADDR
LOAD 1	LD	32+R2
LOAD 2		
LOAD 3		

ROB

	INST	DEST	VAL	
ROB1				
ROB2				
ROB3				
ROB4				
ROB5				
ROB6				

Cycle 1
Issue LD

ARF

Architecture Register File

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	
R2	Read
R3	

RAT

Register Allocation Table

F0	
F1	
F2	
F3	
F4	
F5	
F6	ROB1
F7	
F8	
F9	
F10	

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1						
Add2						
Add3						
Mul1						
Mul2						

Dispatch	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1	2			
LD F2, 45(R3)	1				
MUL F0, F2, F4					
SUB F8, F6, F2					
DIV F10, F0, F6					
ADD F6, F8, F2					

Showing Cycles

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

Load Buffers / RS

	BUSY	ADDR
~~LOAD 1~~	~~LD~~	~~32+R2~~
LOAD 2	LD	45+R3
LOAD 3		

ROB

	INST	DEST	VAL	
ROB1	LD	F6		
ROB2	LD	F2		
ROB3				
ROB4				
ROB5				
ROB6				

Cycle 2
Start LD,
Issue LD

ARF

Architecture Register File

F0	
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	
F9	
F10	
R2	Read
R3	Read

RAT

Register Allocation Table

F0	
F1	
F2	ROB2
F3	
F4	
F5	
F6	ROB1
F7	
F8	
F9	
F10	

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1						
Add2						
Add3						
Mul1						
Mul2						

Dispatch	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1	2	3		
LD F2, 45(R3)	2	3			
MUL F0, F2, F4	3				
SUB F8, F6, F2					
DIV F10, F0, F6					
ADD F6, F8, F2					

Showing Cycles

Load Buffers / RS

	BUSY	ADDR
~~LOAD 1~~	~~LD~~	~~32+R2~~
~~LOAD 2~~	~~LD~~	~~45+R3~~
LOAD 3		

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

ROB

	INST	DEST	VAL	
ROB1	LD	F6		
ROB2	LD	F2		
ROB3	MUL	F0		
ROB4				
ROB5				
ROB6				

Cycle 3
LD1 Complete
LD2 Operands Available
Start EX
Free up RS

ARF

Architecture Register File

F0	
F1	
F2	
F3	
F4	Avalbl
F5	
F6	
F7	
F8	
F9	
F10	
R2	Read
R3	Read

RAT

Register Allocation Table

Connection ARF to RAT

F0	ROB3
F1	
F2	ROB2
F3	
F4	
F5	
F6	ROB1
F7	
F8	
F9	
F10	

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1						
Add2						
Add3						
Mul1	YES	MUL		F4	ROB2	
Mul2						

Dispatch	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1	2	3	4	
LD F2, 45(R3)	2	3	4		
MUL F0, F2, F4	3				
SUB F8, F6, F2	4				
DIV F10, F0, F6					
ADD F6, F8, F2					

Showing Cycles

Load Buffers / RS

	BUSY	ADDR
~~LOAD 1~~	~~LD~~	~~32+R2~~
~~LOAD 2~~	~~LD~~	~~45+R3~~
LOAD 3		

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

ROB

	INST	DEST	VAL	
ROB1	LD	F6	M(A1)	
ROB2	LD	F2		
ROB3	MUL	F0		
ROB4	SUB	F8		
ROB5				
ROB6				

Cycle 4
LD1 writes ROB
Update RAT F6

ARF

Architecture Register File

F0	
F1	
F2	
F3	
F4	Avalbl
F5	
F6	
F7	
F8	
F9	
F10	
R2	Read
R3	Read

RAT

Register Allocation Table

Connection ARF to RAT

F0	ROB3
F1	
F2	ROB2
F3	
F4	
F5	
F6	M(A1)
F7	
F8	ROB4
F9	
F10	

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1	YES	SUB	M(A1)			ROB2
Add2						
Add3						
Mul1	YES	MUL		F4	ROB2	
Mul2						

Dispatch	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1	2	3	4	5
LD F2, 45(R3)	2	3	4	5	
MUL F0, F2, F4	3				
SUB F8, F6, F2	4				
DIV F10, F0, F6	5				
ADD F6, F8, F2					

ROB to ARF

Showing Cycles

Load Buffers / RS

	BUSY	ADDR
~~LOAD 1~~	~~LD~~	~~32+R2~~
~~LOAD 2~~	~~LD~~	~~45+R3~~
LOAD 3		

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

ROB

	INST	DEST	VAL	
~~ROB1~~	~~LD~~	~~F6~~	~~M(A1)~~	
ROB2	LD	F2	M(A2)	
ROB3	MUL	F0		
ROB4	SUB	F8		
ROB5	DIV	F10		
ROB6				

Cycle 5
Commit LD1 F6 to ARF
Free up ROB1
Update RAT Entry F6, F2

ARF

Architecture Register File

F0	
F1	
F2	
F3	
F4	Avalbl
F5	
F6	M(A1)
F7	
F8	
F9	
F10	
R2	Read
R3	Read

RAT

Register Allocation Table

Connection ARF to RAT

F0	ROB3
F1	
F2	M(A2)
F3	
F4	
F5	
F6	~~M(A1)~~
F7	
F8	ROB4
F9	
F10	ROB5

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1	YES	SUB	M(A1)	M(A2)		
Add2						
Add3						
Mul1	YES	MUL	M(A2)	F4	M(A2)	
Mul2	YES	DIV	M(A1)		ROB3	

Dispatch	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1	2	3	4	5
LD F2, 45(R3)	2	3	4	5	6
MUL F0, F2, F4	3	6			
SUB F8, F6, F2	4	6			
DIV F10, F0, F6	5				
ADD F6, F8, F2	6				

ROB to ARF

Showing Cycles

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

Load Buffers / RS

	BUSY	ADDR
~~LOAD 1~~	~~LD~~	~~32+R2~~
~~LOAD 2~~	~~LD~~	~~45+R3~~
LOAD 3		

ROB

	INST	DEST	VAL	
~~ROB1~~	~~LD~~	~~F6~~	~~M(A1)~~	
~~ROB2~~	~~LD~~	~~F2~~	~~M(A2)~~	
ROB3	MUL	F0		
ROB4	SUB	F8		
ROB5	DIV	F10		
ROB6	ADD	F6		

Cycle 6
Commit LD2 F2 to ARF
Free up ROB2
Update RAT Entry F2

ARF

Architecture Register File

F0	
F1	
F2	M(A2)
F3	
F4	Avalbl
F5	
F6	M(A1)
F7	
F8	
F9	
F10	
R2	Read
R3	Read

RAT

Register Allocation Table

F0	ROB3
F1	
F2	~~M(A2)~~
F3	
F4	
F5	
F6	ROB6
F7	
F8	ROB4
F9	
F10	ROB5

Connection ARF to RAT

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
~~Add1~~	~~YES~~	~~SUB~~	~~M(A1)~~	~~M(A2)~~		
Add2	YES	MUL		M(A2)	ROB4	
Add3						
~~Mul1~~	~~YES~~	~~MUL~~	~~M(A2)~~	~~F4~~	~~M(A2)~~	
Mul2	YES	DIV	M(A1)		ROB3	

Dispatch	ISU	EXST	EXCM	WR	COM
LD F6, 32(R2)	1	2	3	4	5
LD F2, 45(R3)	2	3	4	5	6
MUL F0, F2, F4	3	6	15	16	?
SUB F8, F6, F2	4	6	7	8	
DIV F10, F0, F6	5	17	56	57	
ADD F6, F8, F2	6	9	10	11	

ROB to ARF

Showing Cycles

Load Buffers / RS

	BUSY	ADDR
LOAD1		
LOAD2		
LOAD3		

Load=2 Units Delay
Add=2 Units Delay
Mul=10 Units Delay
Div=40 Units Delay

In-order Issue
Single Issue

ROB

	INST	DEST	VAL	
ROB1				
ROB2				
ROB3	MUL	F0		
ROB4	SUB	F8		
ROB5	DIV	F10		
ROB6	ADD	F6		

Cycle ---
Commit LD2 F2 to ARF
Free up ROB2
Update RAT Entry F2

ARF

Architecture Register File

F0	
F1	
F2	M(A2)
F3	
F4	Avalbl
F5	
F6	M(A1)
F7	
F8	
F9	
F10	
R2	Read
R3	Read

RAT

Register Allocation Table

F0	ROB3
F1	
F2	
F3	
F4	
F5	
F6	
F7	
F8	ROB4
F9	
F10	ROB5

Connection ARF to RAT

RS

Add or Sub

Mul or Div

TYPE	BUSY	OP	S1	S2	W1	W2
Add1						
Add2	YES	MUL		M(A2)	ROB4	
Add3						
Mul1	YES	MUL	M(A2)	F4	M(A2)	
Mul2						